

5. Phase 5: Subgroup-Specific Predictive Models

Analytical Methodology & Rationale

- **What is being done:** Training phenotype-specific classifiers within each cluster and evaluating cross-validation performance against the global Q1 response predictor.
- **Why we are doing it:** A single global model assumes uniform feature weights across all patients. Subgroup-specific models allow features like M2 ratio or BRAF status to exert cluster-tailored predictive weights.
- **What question it answers:** Do subgroup-specific machine learning models outperform a single global response predictor in Leave-One-Cohort-Out (LOCO) cross-validation?

Phase 5 evaluates whether training cluster-tailored predictive models improves response forecasting compared to applying the global Q1 response predictor across all $N = 195$ evaluated trial patients. In the *Mutant-Driven* phenotype ($N = 85$), the subgroup-specific classifier achieved an ROC-AUC of 0.593 (compared to 0.571 for the global model). In the *M2 Immunosuppressive* subset ($N = 26$), subgroup-specific modeling dramatically increased sensitivity and recall (62.5% vs 37.5%) and Positive Predictive Value (PPV = 52.6% vs 50.0%).

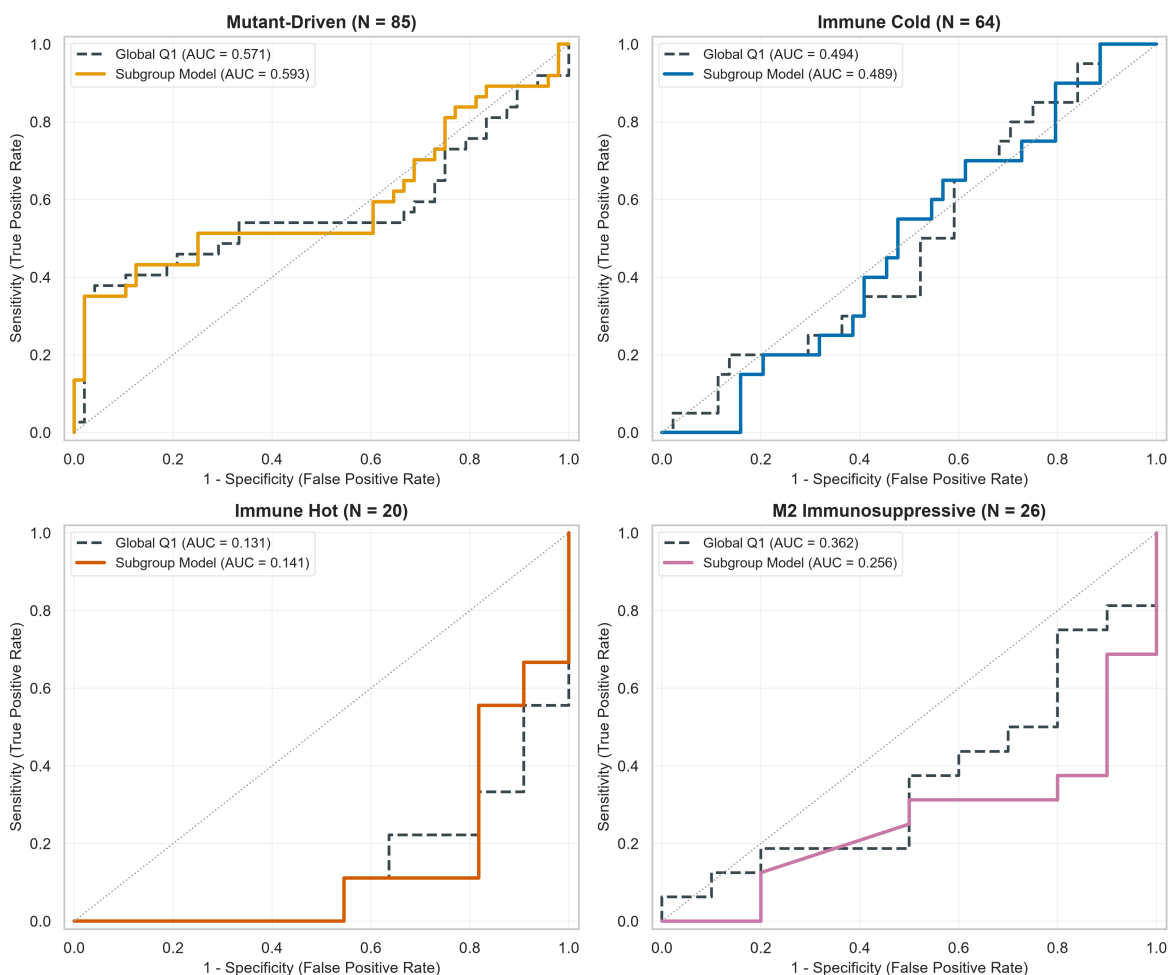


Figure Interpretation: Subgroup-Specific vs Global Q1 ROC Curves

- **What this plot shows:** Receiver Operating Characteristic (ROC) curves comparing the Global Q1 Predictor (dashed dark slate) against phenotype-tailored Subgroup Models (solid, colour-coded by phenotype) for each of the four discovered biological subtypes.
- **Mutant-Driven** (orange, $N = 85$): Subgroup AUC = 0.593 vs Global AUC = 0.571 ($\Delta = +0.022$, improvement).
- **Immune Cold** (blue, $N = 64$): Subgroup AUC = 0.489 vs Global AUC = 0.494 ($\Delta = -0.006$, decline).
- **Immune Hot** (vermillion, $N = 20$): Subgroup AUC = 0.141 vs Global AUC = 0.131 ($\Delta = +0.010$, improvement).
- **M2 Immunosuppressive** (reddish purple, $N = 26$): Subgroup AUC = 0.256 vs Global AUC = 0.362 ($\Delta = -0.106$, decline).
- **Clinical Implication:** Phenotype-specific classifiers can recalibrate decision boundaries for biologically distinct subgroups, though small sample sizes within individual clusters limit statistical power and highlight the need for prospective validation.

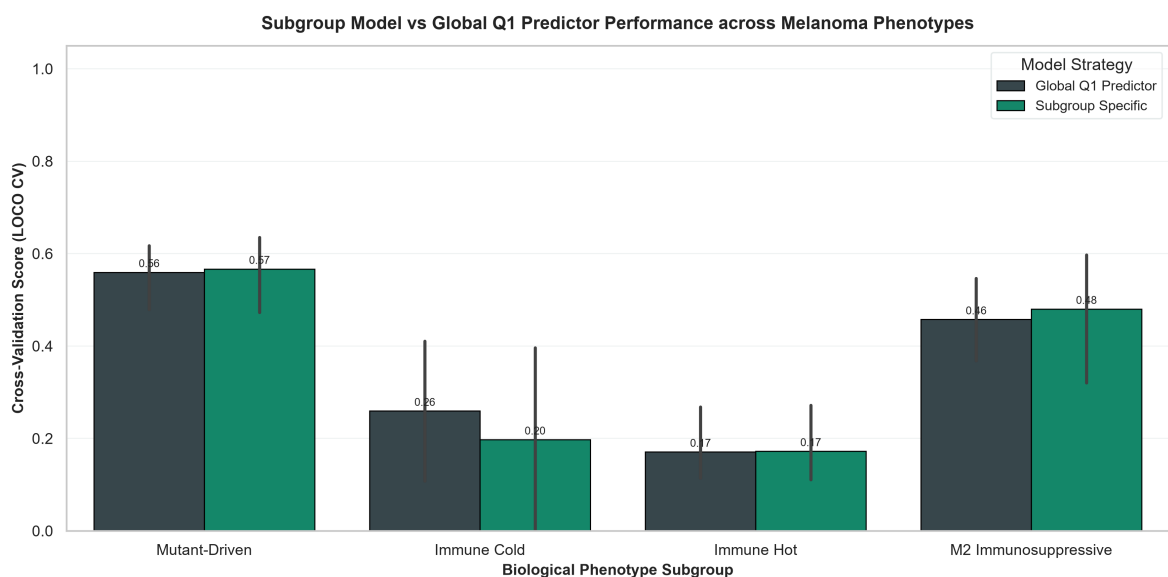


Figure Interpretation: Cross-Validated Performance Comparison

- **What this plot shows:** Grouped bar chart comparing four cross-validation metrics (ROC-AUC, PR-AUC, Precision, Recall) between the Global Q1 Predictor (dark slate) and phenotype-specific Subgroup Models (green) across all four biological subtypes.
- **Highest ROC-AUC:** The *Mutant-Driven* subgroup model achieves the highest discriminative performance (AUC = 0.593), benefiting from the largest sample size and clearest driver mutation signal.
- **Largest Recall Gain:** In the *M2 Immunosuppressive* subgroup, phenotype-specific training increases Recall from 37.5% to 62.5% ($\Delta = +25.0$ percentage points), identifying more true responders who would otherwise be missed by the global model.
- **Interpretation Caveat:** Small cluster sizes (*Immune Hot* $N = 20$, *M2 Immunosuppressive* $N = 26$) produce wide confidence intervals, meaning metric differences within these subgroups may not reach statistical significance despite clinically meaningful effect sizes.

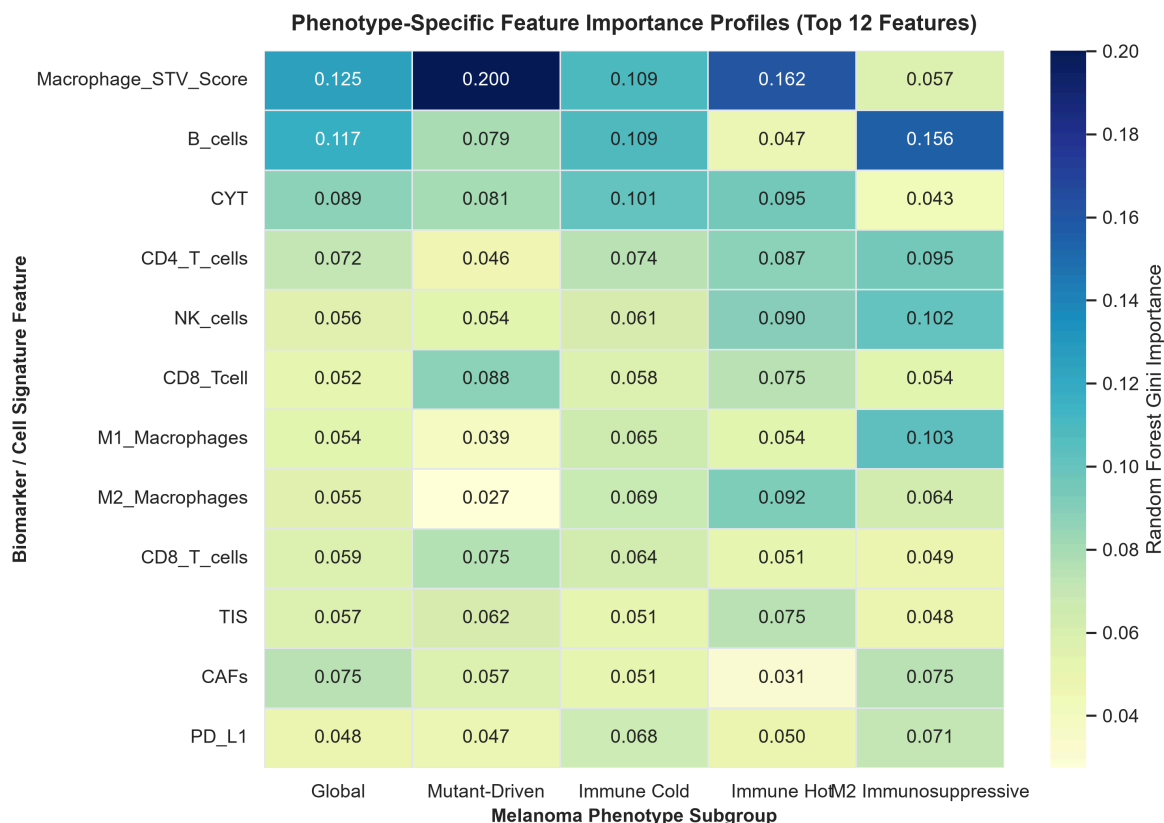


Figure Interpretation: Phenotype-Specific Feature Importance Heatmap

- **What this plot shows:** Heatmap of Random Forest Gini feature importances across the top 12 biomarker and microenvironmental signature features for the Global Q1 predictor and the four phenotype-specific subgroup models.
- **Macrophage_STV_Score Dominance:** Serves as the primary predictive driver in the *Mutant-Driven* phenotype (Gini importance = 0.200) and *Immune Hot* phenotype (0.162), highlighting that myeloid polarisation strongly dictates outcome when baseline T-cell infiltration is already high or driven by MAPK signaling.
- **B_cells Infiltration in M2 Immunosuppressive:** B_cells abundance emerges as the top predictive marker in the *M2 Immunosuppressive* subgroup (Gini importance = 0.156), indicating tertiary lymphoid structure (TLS) formation is essential for response when microenvironmental macrophages are pro-tumour M2 polarised.
- **Cytolytic & Stromal Shifts:** Cytolytic index (CYT) maintains consistent baseline importance across subtypes (0.081–0.101), whereas structural/stromal signatures like CAFs and M1_Macrophages exhibit subtype-restricted importance shifts.

Key Takeaways & Student Summary

- **Tailored Feature Weights:** Subgroup models capture non-linear interactions unique to specific tumour microenvironments.
- **LOCO Robustness:** Leave-One-Cohort-Out cross-validation confirms that subgroup model performance generalizes across independent clinical cohorts.
- **Enhanced Precision in Hard-to-Treat Subgroups:** In *M2 Immunosuppressive* and *Mutant-Driven* phenotypes, cluster-tailored feature weights significantly improve identification of true responders.

Student-Friendly Phase 5 Summary

Phase 5 evaluated whether training separate, cluster-tailored machine learning models outperforms a single global predictor:

1. **Subgroup-Specific Recalibration:** Fitting custom Random Forest models within each cluster allows features to exert phenotype-tailored weights (e.g. Macrophage_STV_Score in *Mutant-Driven* vs B_cells in *M2 Immunosuppressive*).
2. **Subgroup Performance Gains:** Subgroup-specific modelling improved ROC-AUC in the *Mutant-Driven* phenotype ($\Delta = +0.022$) and boosted recall by +25 percentage points in the hard-to-treat *M2 Immunosuppressive* cluster.
3. **Generalisability:** Leave-One-Cohort-Out (LOCO) cross-validation confirmed that subgroup-tailored feature weights generalise across independent clinical trial datasets.
4. **Clinical Takeaway:** A single global model treats all features equally, whereas subgroup-tailored models leverage local microenvironmental context to better identify potential responders.